

## HOPE, BC, Canada

WMO: 711870

Lat: 49.3699N

Lon: 121.4935W

Elev: 39

StdP: 100.86

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 12            | -9.7       | -7.1 | -18.1                         | 0.8 | -7.6 | -14.7 | 1.0 | -5.8 | 10.4                  | 0.8  | 9.6 | -0.1 | 7.2                   | 110  | 0.564 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 30.6            | 19.7 | 28.6 | 19.0 | 26.8 | 18.3 | 20.7                | 28.4 | 19.7 | 27.1 | 18.8 | 25.6 | 4.2                  | 280  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.8                            | 12.8 | 24.2 | 16.8 | 12.1 | 22.8 | 16.0 | 11.4 | 21.6 | 59.9          | 28.7 | 56.4 | 27.2 | 53.4 | 25.8 | 25.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 9.2           | 8.2  | 7.3 | DB                         | -11.4 | 34.5               | 3.2 | 2.3                                                | -13.7 | 36.1       | -15.5 | 37.5       | -17.3 | 38.8       | -19.6 | 40.5 | (4) |
| (5)               |      |     | WB                         | -12.6 | 22.3               | 3.0 | 1.3                                                | -14.8 | 23.2       | -16.5 | 24.0       | -18.2 | 24.7       | -20.4 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.5   | 2.0   | 3.9   | 6.5   | 10.0  | 13.8  | 16.2  | 19.3  | 19.4  | 16.3  | 10.6  | 5.3   | 1.9  |
|                                                                                  | DBStd       | 7.05   | 4.18  | 3.31  | 3.16  | 2.97  | 3.33  | 3.09  | 2.99  | 2.75  | 3.12  | 2.70  | 3.57  | 3.73 |
|                                                                                  | HDD10.0     | 993    | 248   | 169   | 116   | 36    | 5     | 0     | 0     | 0     | 0     | 25    | 143   | 252  |
|                                                                                  | HDD18.3     | 3013   | 506   | 403   | 368   | 251   | 144   | 79    | 22    | 19    | 77    | 241   | 391   | 511  |
|                                                                                  | CDD10.0     | 1164   | 0     | 0     | 6     | 34    | 124   | 186   | 287   | 293   | 189   | 43    | 2     | 0    |
|                                                                                  | CDD18.3     | 142    | 0     | 0     | 0     | 1     | 5     | 16    | 51    | 54    | 16    | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 1467   | 0     | 0     | 1     | 10    | 82    | 187   | 543   | 489   | 155   | 1     | 0     | 0    |
|                                                                                  | CDH26.7     | 430    | 0     | 0     | 0     | 1     | 16    | 56    | 175   | 143   | 38    | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 3.4    | 4.2   | 3.7   | 3.3   | 3.1   | 3.1   | 3.2   | 3.4   | 3.2   | 3.0   | 2.9   | 3.4   | 4.0  |
| Precipitation                                                                    | PrecAvg     | 1565   | 218   | 134   | 171   | 102   | 80    | 78    | 38    | 49    | 83    | 175   | 241   | 206  |
|                                                                                  | PrecMax     | 2493   | 368   | 310   | 331   | 205   | 179   | 151   | 74    | 205   | 188   | 416   | 693   | 336  |
|                                                                                  | PrecMin     | 1219   | 127   | 7     | 17    | 37    | 21    | 0     | 3     | 0     | 0     | 34    | 72    | 104  |
|                                                                                  | PrecStd     | 260    | 71    | 74    | 82    | 43    | 36    | 31    | 21    | 45    | 51    | 92    | 124   | 62   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 11.1  | 12.6  | 19.4  | 24.6  | 28.8  | 31.6  | 34.1  | 32.6  | 30.2  | 21.1  | 13.6  | 10.3 |
|                                                                                  |             | MCWB   | 8.6   | 7.9   | 11.0  | 14.7  | 17.0  | 20.0  | 20.6  | 19.4  | 18.2  | 14.3  | 11.6  | 8.2  |
|                                                                                  | 2%          | DB     | 9.1   | 10.6  | 15.5  | 21.0  | 25.6  | 28.2  | 30.7  | 30.1  | 27.3  | 18.1  | 11.9  | 8.4  |
|                                                                                  |             | MCWB   | 7.3   | 7.2   | 9.5   | 12.8  | 16.2  | 18.4  | 20.4  | 19.6  | 17.5  | 12.6  | 10.2  | 6.9  |
|                                                                                  | 5%          | DB     | 7.5   | 9.3   | 13.0  | 18.2  | 23.0  | 25.1  | 28.4  | 28.1  | 24.8  | 16.0  | 10.6  | 7.1  |
|                                                                                  |             | MCWB   | 6.0   | 6.7   | 8.3   | 11.3  | 15.3  | 17.4  | 19.6  | 19.2  | 17.1  | 12.3  | 9.2   | 5.7  |
|                                                                                  | 10%         | DB     | 6.4   | 8.1   | 11.0  | 15.6  | 20.5  | 22.7  | 26.3  | 26.2  | 22.3  | 14.4  | 9.5   | 6.1  |
|                                                                                  |             | MCWB   | 5.1   | 6.0   | 7.6   | 10.1  | 14.1  | 16.0  | 18.7  | 18.6  | 15.9  | 11.7  | 8.3   | 4.8  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 9.6   | 9.6   | 12.0  | 15.5  | 18.2  | 20.7  | 22.8  | 21.6  | 19.8  | 16.0  | 12.4  | 8.9  |
|                                                                                  |             | MCDB   | 10.4  | 10.7  | 17.1  | 23.2  | 26.4  | 29.7  | 31.4  | 29.3  | 27.4  | 18.9  | 13.4  | 9.7  |
|                                                                                  | 2%          | WB     | 7.5   | 8.3   | 10.4  | 13.3  | 16.9  | 18.9  | 21.1  | 20.5  | 18.6  | 14.2  | 10.7  | 7.2  |
|                                                                                  |             | MCDB   | 8.7   | 9.6   | 14.0  | 19.3  | 24.0  | 26.7  | 29.1  | 28.0  | 25.2  | 16.2  | 11.6  | 8.1  |
|                                                                                  | 5%          | WB     | 6.4   | 7.2   | 9.1   | 11.9  | 15.7  | 17.5  | 19.9  | 19.6  | 17.5  | 13.0  | 9.6   | 6.1  |
|                                                                                  |             | MCDB   | 7.3   | 8.8   | 11.9  | 17.0  | 21.9  | 24.3  | 27.2  | 26.8  | 23.3  | 15.0  | 10.4  | 6.7  |
|                                                                                  | 10%         | WB     | 5.4   | 6.2   | 8.1   | 10.6  | 14.5  | 16.3  | 18.9  | 18.7  | 16.4  | 12.0  | 8.6   | 5.1  |
|                                                                                  |             | MCDB   | 6.3   | 7.8   | 10.5  | 14.6  | 19.7  | 21.8  | 25.6  | 25.1  | 21.1  | 13.9  | 9.3   | 5.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 4.2   | 6.0   | 7.6   | 9.2   | 9.6   | 8.9   | 10.5  | 10.5  | 9.6   | 6.6   | 4.3   | 3.7  |
|                                                                                  |             | MCDBR  | 5.4   | 7.7   | 11.8  | 14.5  | 14.8  | 14.7  | 14.7  | 14.5  | 9.5   | 5.3   | 4.8   |      |
|                                                                                  |             | MCWBR  | 4.0   | 4.9   | 6.5   | 7.7   | 7.4   | 7.3   | 6.8   | 6.5   | 6.9   | 5.4   | 4.1   | 3.8  |
|                                                                                  | 5% WB       | MCDBR  | 5.1   | 6.4   | 9.7   | 12.7  | 13.1  | 13.4  | 13.5  | 13.1  | 12.8  | 7.7   | 5.0   | 4.7  |
| MCWBR                                                                            |             | 4.2    | 4.6   | 5.9   | 7.0   | 6.8   | 6.9   | 6.6   | 6.2   | 6.4   | 4.9   | 4.2   | 4.0   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.299  | 0.299 | 0.316 | 0.361 | 0.399 | 0.377 | 0.365 | 0.370 | 0.353 | 0.365 | 0.315 | 0.293 |      |
|                                                                                  | taud        | 2.306  | 2.332 | 2.310 | 2.246 | 2.282 | 2.388 | 2.443 | 2.456 | 2.487 | 2.387 | 2.442 | 2.344 |      |
|                                                                                  | Ebn at Noon | 744    | 849   | 894   | 883   | 861   | 882   | 888   | 868   | 848   | 758   | 730   | 700   |      |
|                                                                                  | Edh at Noon | 70     | 86    | 105   | 125   | 127   | 115   | 107   | 101   | 89    | 83    | 62    | 59    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.06   | 1.92  | 3.01  | 4.51  | 5.32  | 5.60  | 6.47  | 5.52  | 3.84  | 2.21  | 1.09  | 0.82  |      |
|                                                                                  | RadStd      | 0.14   | 0.19  | 0.31  | 0.31  | 0.46  | 0.58  | 0.58  | 0.39  | 0.34  | 0.25  | 0.12  | 0.10  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +89     | +32     |
| (47) Regional (58 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page